SPA617 Radio Astronomy Laboratory

Experiment 1: RF Transmission, Reflections and Standing Waves

Arunkumar Kotagi
261310604

19/08/2026

1 Aim

The aim of this experiment is to study signal propagation and reflection in a RF transmission line.

2 Objectives

(a) To find the phase velocity and velocity factor of the signal in the transmission cable.

(b) To study the reflection from an open-ended cable and from a cable terminated with a $50\,\Omega$ load.

(c) To estimate the magnitude of the reflection coefficient and check whether the cable is approximately matched to $50\,\Omega$.

(d) To calculate the inductance and capacitance per unit length of the cable.

(e) To measure the insertion loss and attenuation per unit length of the cable.

3 Theory

A transmission line is used to guide electromagnetic energy. When the load impedance is different from the characteristic impedance of the cable, part of the signal is reflected back towards the source. At the Tee junction, the incident and reflected signals add together.

When the two signals are in phase, constructive interference occurs and the measured voltage is maximum. When they are out of phase by 180°, destructive interference occurs

and the measured voltage is minimum. Therefore, while the frequency is changed, the measured voltage at the Tee junction changes periodically.

For a cable of length ℓ, the reflected signal travels a round-trip distance 2ℓ. If two maxima or two minima are separated by Δf, the round-trip time is

$$\tau = \frac{1}{\Delta f}. \tag{1}$$

then the phase velocity is

$$v_p = 2\ell\Delta f, \tag{2}$$

and the velocity factor is

$$VF = \frac{v_p}{c}, \tag{3}$$

and $c \simeq 3 \times 10^8\,\mathrm{m/s}$ is the speed of light in vacuum.

The voltage reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \tag{4}$$

For the measured maximum and minimum voltages, its magnitude can be estimated by

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{5}$$

For a $50\,\Omega$ matched load, the reflection should be much smaller than that of an open circuit. Thus, a large reduction in the reflection coefficient with the $50\,\Omega$ termination shows that the cable is approximately matched to $50\,\Omega$.

For a low-loss transmission line,

$$v_p = \frac{1}{\sqrt{L'C'}}, \qquad Z_0 = \sqrt{\frac{L'}{C'}}. \tag{6}$$

Therefore,

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p}. \tag{7}$$

The insertion loss of the cable is

$$IL = P_{\mathrm{ref}} - P_{\mathrm{cable}}, \tag{8}$$

and the attenuation per unit length is

$$\alpha = \frac{IL}{\ell} \qquad \text{dB/m.} \tag{9}$$

4 Apparatus

The following apparatus was used:

- RF signal generator
- Digital storage oscilloscope
- Spectrum analyser
- RF Tee connector
- Transmission cables
- $50\,\Omega$ termination
- RF adapters and connectors

5 Experimental Setup and Procedure

The signal generator to tee connector then one end of the tee connector is connceted to the oscilloscope and the other end of the tee connector was connected with open transmission cable. The length of the transmission cable which was under test is measured.

$$\ell = 91.5\,\text{cm} = 0.915\,\text{m}. \tag{10}$$

For Part A, the far end of the cable was left open. The frequency was slowly changed and the RMS voltage at the Tee junction was recorded.

For Part B, the same setup was used first with the cable open and then with a $50\,\Omega$ load connected at the far end. The maximum and minimum RMS voltages were recorded.

For Part C, the phase velocity found from Part A and the approximate value $Z_0 \simeq 50\,\Omega$ were used to calculate L' and C'.

For Part D, a short reference connection was first used to measure the reference power. The cable under test was then inserted and the cable power was measured using the spectrum analyser.

6 Part A: Phase Velocity and Velocity Factor

6.1 Measured Data

Table 1: Measured V_{RMS} as a function of frequency for the open-ended cable.

f (MHz)	V_{RMS} (V)	**Observation**	f (MHz)	V_{RMS} (V)	**Observation**
10.0	1.21		35.0	1.75	
11.0	1.06		36.0	1.71	
12.0	0.88168		37.0	1.65	
13.0	0.72045		38.0	1.59	
14.0	0.55041		39.0	1.54	
15.0	0.38560		40.0	1.48	
16.0	0.21936		41.0	1.40	
17.0	0.08470		42.0	1.30	
17.2	0.07608	Minimum	43.0	1.19	
17.5	0.09190		44.0	1.07	
18.0	0.15607		45.0	0.9601	
19.0	0.30907		46.0	0.8545	
20.0	0.46867		47.0	0.7366	
21.0	0.62676		48.0	0.612	
22.0	0.76622		49.0	0.490	
23.0	0.90745		50.0	0.335	
24.0	1.05		51.0	0.207	
25.0	1.16		52.0	0.120	Minimum
26.0	1.30		53.0	0.187	
27.0	1.40		54.0	0.322	
28.0	1.47		55.0	0.490	
29.0	1.50		56.0	0.640	
30.0	1.63		57.0	0.795	
31.0	1.68		58.0	0.948	
32.0	1.73		59.0	1.08	
33.0	1.77		60.0	1.20	
34.0	1.78	Maximum			

6.2 Standing-Wave Plot

Figure 1 shows all the measured points. The clear consecutive minima used for the phase-velocity calculation are at 17.2 MHz and 52.0 MHz. These points are marked on the graph.

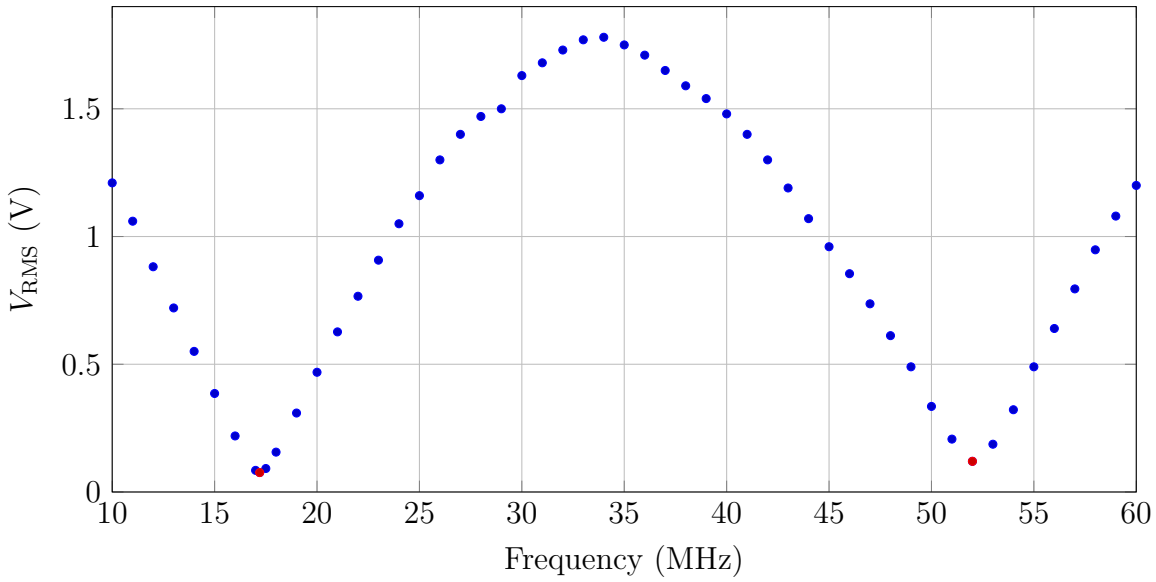


Figure 1: Measured RMS voltage versus frequency for the open-ended cable.

6.3 Calculation of Frequency Spacing

Using the two consecutive minima,

$$f_1 = 17.2\,\mathrm{MHz}, \tag{11}$$
$$f_2 = 52.0\,\mathrm{MHz}. \tag{12}$$

as

$$\begin{aligned} \Delta f &= |f_2 - f_1| && (13)\\ &= |52.0 - 17.2|\,\mathrm{MHz} && (14)\\ &= 34.8\,\mathrm{MHz} && (15)\\ &= 3.48 \times 10^7\,\mathrm{Hz}. && (16) \end{aligned}$$

Using $\ell = 0.915\,\mathrm{m}$,

$$\begin{aligned} v_p &= 2\ell\Delta f && (17)\\ &= 2(0.915)(3.48 \times 10^7) && (18)\\ &= 6.3684 \times 10^7\,\mathrm{m/s}. && (19) \end{aligned}$$

so

$$VF = \frac{v_p}{c} \tag{20}$$

$$= \frac{6.3684 \times 10^7}{3.0 \times 10^8} \tag{21}$$

$$= 0.2123. \tag{22}$$

Thus the measured phase velocity is approximately

$$\boxed{v_p = 6.37 \times 10^7 \,\mathrm{m/s}}$$

and the velocity factor is

$$\boxed{VF = 0.212}.$$

7 Part B: Reflection Coefficient and Characteristic Impedance

The measured values are

Table 2: Maximum and minimum RMS voltages for the two terminations.

Termination	$V_{\max}$ (V)	$V_{\min}$ (V)	$\lvert\Gamma\rvert$
Open circuit	1.72	0.0453	0.9487
50 Ω load	1.06	0.93336	0.0635

For the open circuit,

$$|\Gamma_{\text{open}}| = \frac{1.72 - 0.0453}{1.72 + 0.0453} \tag{23}$$

$$= 0.9487. \tag{24}$$

For the 50 Ω load,

$$|\Gamma_{50}| = \frac{1.06 - 0.93336}{1.06 + 0.93336} \tag{25}$$

$$= 0.0635. \tag{26}$$

The reflection coefficient is much smaller for the 50 Ω termination

$$|\Gamma_{\text{open}}| \approx 0.949, \qquad |\Gamma_{50}| \approx 0.064.$$

This shows that the $50\,\Omega$ termination reduces the reflected signal strongly. Hence, the cable is approximately matched to $50\,\Omega$

8 Part C: Inductance and Capacitance per Unit Length

Using

$$Z_0 \simeq 50\,\Omega$$

and

$$v_p = 6.3684 \times 10^7\,\text{m/s},$$

the inductance per unit length is

$$\begin{aligned} L' &= \frac{Z_0}{v_p} && (27)\\ &= \frac{50}{6.3684 \times 10^7} && (28)\\ &= 7.8513 \times 10^{-7}\,\text{H/m}. && (29) \end{aligned}$$

Therefore,

$$\boxed{L' = 785.1\,\text{nH/m}}.$$

The capacitance per unit length is

$$\begin{aligned} C' &= \frac{1}{Z_0 v_p} && (30)\\ &= \frac{1}{50(6.3684 \times 10^7)} && (31)\\ &= 3.1405 \times 10^{-10}\,\text{F/m}. && (32) \end{aligned}$$

Therefore,

$$\boxed{C' = 314.1\,\text{pF/m}}.$$

9 Part D: Cable Attenuation

The cable length used was

$$\ell = 0.915\,\text{m}.$$

The given reference power is $-1.4\,\text{dBm}$. The power reference is frequency dependent variable it varies slightly for all higher frequency. But we are consider it same for simplicity of the experiment.

Table 3: Measured cable power and calculated attenuation.

Frequency (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	α (dB/m)
100	-1.40	-1.80	0.437
150	-1.40	-1.96	0.612
250	-1.40	-2.12	0.787
300	-1.40	-2.18	0.852

At $100\,\text{MHz}$,

$$\begin{aligned} IL &= P_{\text{ref}} - P_{\text{cable}} && (33)\\ &= (-1.40) - (-1.80) && (34)\\ &= 0.40\,\text{dB}. && (35) \end{aligned}$$

Thus,

$$\begin{aligned} \alpha &= \frac{0.40}{0.915} && (36)\\ &= 0.437\,\text{dB/m}. && (37) \end{aligned}$$

The same calculation was made at all the measured frequencies.

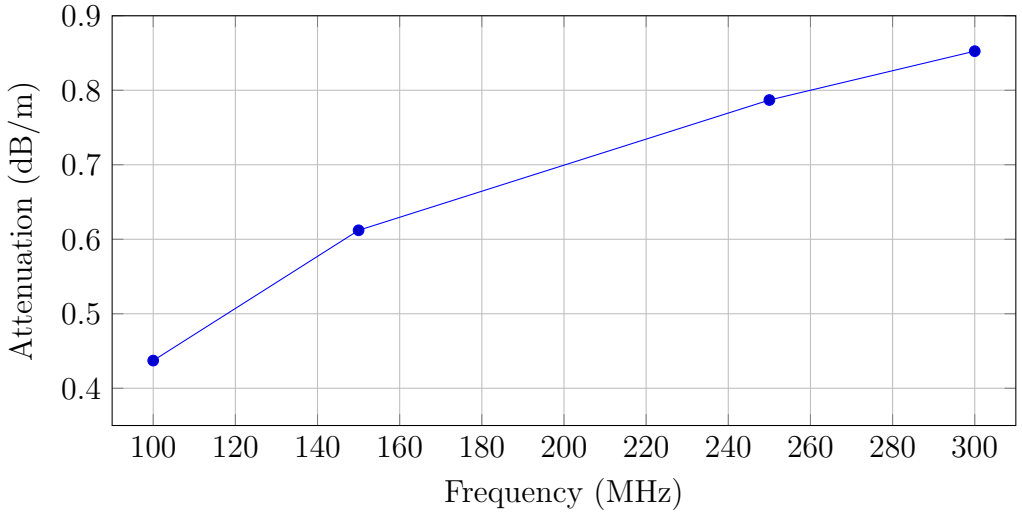


Figure 2: Attenuation per unit length as a function of frequency.

The plot shows that the cable attenuation increases with frequency for the measured range.

10 Results

Table 4: Summary of measured and calculated results.

Quantity	**Value**
Cable length, ℓ	0.915 m
Frequency spacing, Δf	34.8 MHz
Phase velocity, v_p	6.37×10^7 m/s
Velocity factor, VF	0.212
Approximate characteristic impedance, Z_0	50 Ω
Open-circuit reflection coefficient, $\lvert\Gamma_{\text{open}}\rvert$	0.949
50 Ω reflection coefficient, $\lvert\Gamma_{50}\rvert$	0.0635
Inductance per unit length, L'	785.1 nH/m
Capacitance per unit length, C'	314.1 pF/m
Attenuation at 100 MHz	0.437 dB/m
Attenuation at 300 MHz	0.852 dB/m

11 Conclusion

We studied and observed the characteristics of the RF transmission line by measuring the voltage, which gave us the minima and maxima after the reflection. We calculated the Velocity factor, reflection coefficient, inductance, capacitance and attenuation of the the transmission cable under test

11.1 Key takeaways

- I came to know that we measure the V_{rms} at tee connector.
- When we short the transmission cable, the reflected wave will 180° phase shift, which will cause a destructive interference.
- Any unmatched impedance at the open end of the transmission line give heavy reflections.
- We learnt that the attenuation in the transmission cable is frequency dependent, which implies that we need to take care of the losses for every measurement.

AI usage : Used chatgpt to plot the graphs and formatting the latex file.